FIFA Rankings vs ELO Ratings: Predictive Validity in World Cup Knockout Stages (1994–2022)

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Abstract

Football's most consequential matches — World Cup knockout encounters — are seeded and drawn using FIFA World Rankings. Yet a rival system, ELO ratings, has long been considered theoretically superior. This study asks a direct question: which system actually predicts knockout outcomes better, and did FIFA's 2018 decision to adopt an ELO-based formula close the gap?

We analysed 115 knockout matches across eight World Cups from 1994 to 2022, fitting separate logistic regression models for each rating system — specified without an intercept, consistent with the neutral-venue setting — and evaluating them across six metrics: prediction accuracy, discrimination (AUC-ROC), calibration (Brier score), goal difference explained (OLS R^2), out-of-sample cross-validation, and controlled regression (McFadden's pseudo-R^2 against a 50/50 null). ELO outperformed FIFA World Rankings on every single measure. It correctly predicted knockout winners 73.9% of the time versus FIFA's 68.7%, achieved an AUC of 0.775 versus 0.695, and produced better-calibrated probability estimates (Brier score 0.189 versus 0.220).

The 2018 FIFA formula change genuinely improved FIFA's predictive validity — the accuracy gap narrowed from 6.0 percentage points to 3.2. It did not close it. A formal McNemar's test ($p = 0.180$) could not confirm the accuracy difference at conventional significance thresholds, reflecting a structural data constraint: achieving 80% power requires approximately 88 discordant pairs, equivalent to roughly 506 total knockout matches. The convergent direction of evidence across six complementary metrics, combined with ELO exceeding FIFA in 89.8% of 10,000 bootstrap resamples, supports the conclusion that ELO ratings are more predictively valid. Three robustness checks confirm the performance gap is structural — not an artefact of stale ranking data, not driven by the 1994 tournament, and not resolved by substituting FIFA points for rank.

Keywords: ELO ratings; FIFA World Rankings; football prediction; World Cup; knockout stage; logistic regression; calibration; McNemar's test; predictive validity; staleness analysis

Contents

1 Introduction

Football ranking systems serve two practical functions: seeding tournament draws and measuring national team prestige. FIFA World Rankings have performed both since 1992, officially determining which teams occupy favoured positions at every World Cup draw. The system has attracted sustained criticism from analysts who argue that ELO ratings — a continuous, opponent-weighted system originally designed for chess — provide a more accurate picture of team quality.

The stakes are not trivial. If FIFA rankings systematically misidentify the stronger team, tournament draws are distorted, and the teams most likely to win are not adequately protected from early collisions. For broadcasters, analysts, and betting markets, the question of which system better predicts outcomes has direct practical value. For FIFA itself, the question carries institutional weight — the organisation spent over two decades defending a formula that produced 37.5% accuracy at the 2006 World Cup — a single-tournament result that, while striking, should be interpreted cautiously given the small sample of 16 matches.

The comparison between ELO and FIFA rankings is not new. Hvattum and Arntzen (2010) found ELO outperformed simple benchmark methods. Lasek, Szlávik, and Bhulai (2013) confirmed ELO-based approaches consistently outperformed the official FIFA ranking. Wunderlich and Memmert (2016) found persistent FIFA inferiority to betting markets even after the 2006 formula revision.

This paper addresses three gaps in the existing literature. First, it isolates knockout stage matches specifically, where seeding consequences are most severe and squad rotation is absent. Second, it exploits the 2018 formula change as a pre/post comparison to assess whether FIFA's adoption of ELO-based logic narrowed the predictive gap. Third, it applies novel robustness checks using real FIFA ranking snapshot dates, a sensitivity test excluding the 1994 tournament, and a comparison against FIFA points. Three research questions guide the analysis: Does ELO outperform FIFA in predicting World

Cup knockout outcomes across 1994–2022? Has predictive power changed across the 2018 formula change? Does ranking staleness materially affect the conclusions?

1.1 Hypotheses

Based on prior literature and the theoretical properties of each system, three directional hypotheses are advanced:

H1 ELO ratings have greater predictive validity than FIFA World Rankings for World Cup knockout stage outcomes across 1994–2022, as measured by prediction accuracy, AUC-ROC, Brier score, goal-difference R^2, out-of-sample cross-validation accuracy, and controlled regression (McFadden's pseudo-R^2).

H2 FIFA's 2018 adoption of an ELO-based formula improved its predictive validity, narrowing — but not closing — the performance gap relative to ELO ratings.

H3 Ranking staleness does not materially account for FIFA's predictive inferiority; the gap is structural and persists after controlling for the number of days between the ranking snapshot and the match date.

2 Background and Related Literature

2.1 The ELO Rating System

The ELO system was developed by Árpád Élő for chess (Élő, 1978) and adapted for team sports. The update rule is:

$$R_{\text{new}} = R_{\text{old}} + K \cdot G \cdot (W - W_e),$$

where R is the team's rating, K is a match importance weight, G is a goal difference multiplier, W is the match outcome (1 = win, 0.5 = draw, 0 = loss), and

$$W_e = \frac{1}{1 + 10^{-\Delta r/400}}$$

is the expected outcome derived from the pre-match rating difference Δr. The eloratings.net implementation applies no home advantage correction for neutral venues, making ELO ratings fully appropriate for this comparison (eloratings.net, 2024).

2.2 FIFA World Rankings: Pre-2018 Formula

FIFA introduced its World Rankings in 1992 using a points-based averaging system over an eight-year rolling window. Two fundamental flaws have been documented in the literature (Lasek et al., 2013): the system rewarded match volume over quality, and goal difference was entirely ignored. These design choices made the formula susceptible to manipulation through friendly match scheduling. The consequences were most visible at the 2006 World Cup, where FIFA accuracy in this study's knockout analysis fell to 37.5% — well below the 50% chance threshold.

2.3 FIFA World Rankings: Post-2018 Formula

In August 2018, FIFA replaced its methodology with a system structurally based on ELO (Fédération Internationale de Football Association, 2018). Two implementation differences from eloratings.net remain: FIFA does not weight outcomes by goal difference, and uses a divisor of 600 rather than the standard 400, making probability estimates less sensitive to rating gaps. Both are reversible policy choices that this study's evidence suggests should be revisited.

2.4 Prior Evidence on Predictive Validity

Hvattum and Arntzen (2010) found ELO outperformed simpler benchmarks in the English Premier League, though it was outperformed by betting odds. Lasek et al. (2013) confirmed ELO-based approaches outperformed FIFA rankings, noting friendly match scheduling as a significant source of predictive invalidity. Wunderlich and Memmert (2016) found improvement after the 2006 formula revision but persistent inferiority to betting markets. No prior study has isolated knockout stages, spanned the 2018 formula change as a pre/post comparison, or applied real-data staleness controls using verified FIFA snapshot dates.

3 Data and Methodology

3.1 Dataset Construction

The analysis covers eight World Cup knockout stages (USA 1994 through Qatar 2022). Group stage matches were excluded due to squad rotation and differential qualification stakes. Four data sources were merged: match results from Badrnezhad (2024); ELO ratings from eloratings.net (2024), validated match-by-match against saved source pages; FIFA rankings, points, and snapshot dates from Alex (2024); and squad data from Fjelstul (2023).

The merged dataset contains 128 knockout matches. After removing 13 matches where ELO or FIFA ranking data could not be reliably matched, the analysis dataset contains 115 matches. Matches decided by penalty shootouts are treated as wins for the shootout winner in accuracy analysis but excluded from goal difference regression.

3.2 Variables

Dependent variable. The primary outcome is a binary indicator of whether the home-listed team won the match. Since all knockout matches are at neutral venues, the home/away designation carries no competitive significance — it reflects only administrative draw order.

Independent variables. $\Delta\text{ELO} = \text{ELO}_{\text{home}} - \text{ELO}_{\text{away}}$ (mean = 55.0, SD = 142.1); $\Delta\text{FIFA} = \text{FIFA\_rank}_{\text{away}} - \text{FIFA\_rank}_{\text{home}}$ (mean = 3.3, SD = 16.7). Both are signed so positive values indicate the home-listed team is stronger. FIFA rank is the primary FIFA predictor because it is the actual seeding instrument FIFA uses for World Cup draws.

Staleness variable. *Days_stale* measures the number of days between the FIFA ranking snapshot date and the match date, using real `rank_date` values from the FIFA ranking dataset (Alex, 2024) — not proxy estimates. Real staleness ranged from 0 to 73 days (mean 35.5 days).

Figure 1 shows the distributions of ELO differences, FIFA rank differences, and ranking staleness across the 115 matches.

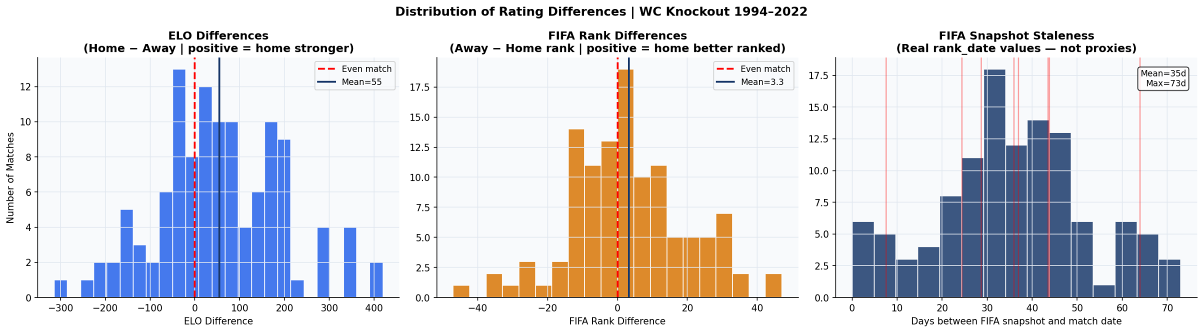

Figure 1: Distribution of rating differences and FIFA snapshot staleness across 115 World Cup knockout matches, 1994–2022. Left: ELO difference (home minus away; positive = home stronger). Centre: FIFA rank difference (away minus home rank; positive = home better ranked). Right: days between FIFA ranking snapshot and match date using real `rank_date` values. Mean staleness = 35 days; maximum = 73 days.

3.3 Statistical Methods

3.3.1 Logistic Regression

Separate logistic regression models without an intercept are fitted for ELO and FIFA using the `statsmodels` library (Seabold & Perktold, 2010), with a no-intercept specification required by the neutral-venue setting: equal teams at a neutral ground should have exactly a 50% win probability.

3.3.2 McNemar's Test

McNemar's test (McNemar, 1947) is applied to formally test the accuracy difference:

$$\chi^2 = \frac{(b-c)^2}{b+c},$$

where b = matches correctly predicted by ELO but not FIFA ($b = 13$), and c = the reverse ($c = 7$). The test operates only on discordant pairs ($n_d = 20$).

3.3.3 Bootstrap Confidence Intervals

A paired percentile bootstrap with 10,000 resamples is applied. Each resample draws the same matched rows for both systems simultaneously, preserving the within-match pairing. Sensitivity analysis presents confidence intervals at 90%, 95%, and 99% levels.

3.3.4 Calibration Analysis

The Brier score (Brier, 1950) summarises calibration:

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2.$$

A random coin-flip yields BS = 0.250; perfect predictions yield BS = 0.000.

3.3.5 Cross-Validation

Leave-one-tournament-out cross-validation evaluates out-of-sample accuracy across eight tournament folds. For each fold, the held-out tournament's outcomes are predicted using the rating differences alone — no model training on the test fold.

3.3.6 Controlled Logistic Regression

A final specification adds round (ordinal), centred year, and a post-2018 dummy to isolate the pure rating signal from structural variation across rounds and eras.

3.4 Statistical Power

With $n_d = 20$ discordant pairs and $p_1 = 0.65$, McNemar's power is:

$$\text{power} = \Phi\left(\frac{0.65 - 0.50}{\sqrt{0.25/20}} - 1.96\right) \approx 0.268.$$

Achieving 80% power requires approximately 88 discordant pairs, equivalent to roughly 506 total knockout matches. See Appendix B for the full derivation.

4 Results

4.1 Descriptive Statistics

Table 1 reports descriptive statistics. The two systems agree on the predicted stronger team in 82.6% of the 115 matches; the remaining 20 constitute the discordant pairs for McNemar's test.

Table 1: Descriptive statistics for key variables (N = 115 World Cup knockout matches, 1994–2022). Staleness computed from real FIFA `rank_date` values.

Variable	N	Mean	SD	Min	Max
ELO difference (Δ_{ELO})	115	55.0	142.1	−314	420
FIFA rank difference (Δ_{FIFA})	115	3.3	16.7	−47	47
Goal difference	89	0.3	1.9	−6	5
Days stale (real `rank_date`)	115	35.5	16.8	0	73
ELO–FIFA agreement on favourite	95	82.6%			

4.2 Logistic Regression and Discrimination

Table 2 presents the baseline logistic regression results. Both rating difference variables are statistically significant predictors (ELO: $\beta = 0.0084$, $SE = 0.0019$, $p < 0.001$; FIFA: $\beta = 0.0460$, $SE = 0.0136$, $p = 0.001$). ELO achieves prediction accuracy of 73.9% versus FIFA's 68.7%, and an AUC-ROC of 0.775 versus 0.695. McFadden's pseudo-R^2 for the ELO model (0.188) is 2.1 times that of the FIFA model (0.089), computed against a 50/50 null. Figure 2 presents the ROC curves.

Table 2: Baseline logistic regression results (no-intercept specification, $N = 115$). Pseudo-R^2 computed against a 50/50 null consistent with neutral-venue assumption.

	ELO Model		**FIFA Model**	
	Estimate	p-value	Estimate	p-value
Coefficient (β)	0.0084	< 0.001	0.0460	0.001
SE	0.0019		0.0136	
95% CI	[0.0047, 0.0121]		[0.0194, 0.0727]	
Accuracy	73.9%		68.7%	
AUC-ROC	0.775		0.695	
Pseudo-R^2 (50/50 null)	0.188		0.089	
Log-likelihood	-64.691		-72.652	

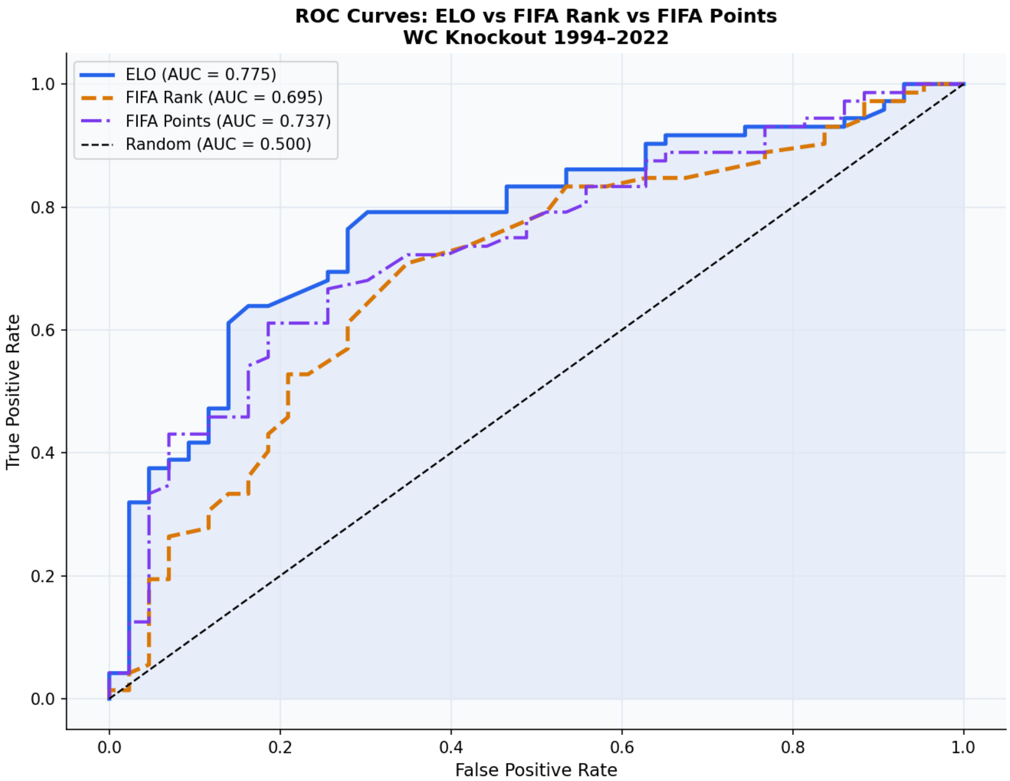

Figure 2: ROC curves for ELO ratings (AUC = 0.775), FIFA rank (AUC = 0.695), and FIFA points (AUC = 0.737) predicting World Cup knockout outcomes, 1994–2022 ($N = 115$). The random classifier baseline (AUC = 0.500) is shown as a dashed diagonal. ELO dominates both FIFA predictors at all threshold values.

4.3 McNemar's Test and Statistical Power

The McNemar's test yields $\chi^2 = 1.800$, $p = 0.180$ ($b = 13$, $c = 7$, $n_d = 20$, ratio $b/c = 1.86$). The test fails to reject the null hypothesis of equal accuracy at $\alpha = 0.05$. Cohen's $h = 0.115$ (small effect).

Statistical power is 26.8% — the test would fail to detect ELO's superiority 73.2% of the time even if genuine. Achieving 80% power requires approximately 88 discordant pairs, implying roughly 506 total knockout matches. The non-significance reflects a structural data constraint, not evidence of equivalence. Figure 9 shows the power curve.

4.4 Bootstrap Confidence Intervals

Bootstrap resampling (10,000 paired iterations) yields: ELO accuracy 73.9% [95% CI: −1.8%, +13.0% for the gap]; FIFA accuracy 68.7%; accuracy gap +5.2 percentage points. ELO exceeded FIFA in **89.8%** of all 10,000 bootstrap resamples.

The multi-level sensitivity analysis (Figure 3) shows the gap CI includes zero at all three thresholds (90%: [−0.9%, +11.3%]; 95%: [−1.8%, +13.0%]; 99%: [−4.3%, +15.7%]), consistent with the power constraint. The 89.8% bootstrap superiority rate is the most informative single statistic.

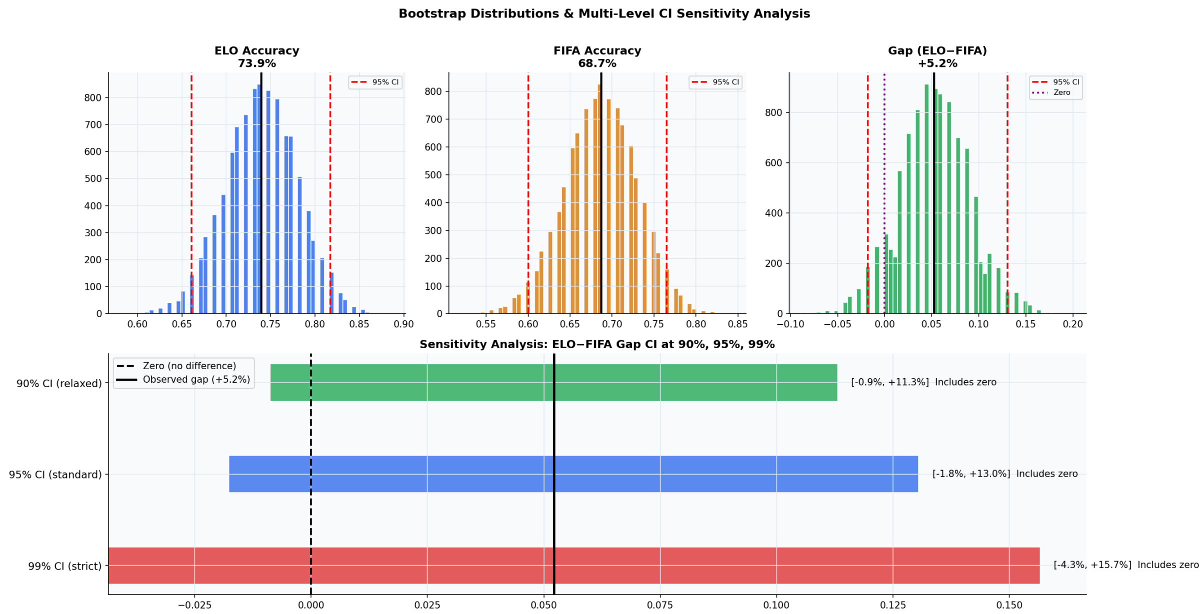

Figure 3: Bootstrap distributions and multi-level confidence interval sensitivity analysis. Top row: bootstrap distributions of ELO accuracy (73.9%), FIFA accuracy (68.7%), and the gap (ELO−FIFA, +5.2 pp) from 10,000 paired resamples. Bottom panel: the accuracy gap CI at 90%, 95%, and 99% levels, all including zero — consistent with 26.8% statistical power. ELO exceeded FIFA in 89.8% of bootstrap resamples.

4.5 Calibration

Table 3 presents calibration metrics. ELO achieves a Brier score of 0.1892 — a 24.3% improvement over the random baseline. FIFA achieves BS = 0.2200, a 12.0% improvement. Figure 4 shows FIFA's calibration curve exhibits systematic miscalibration in the 0.40–0.55 probability range — a direct consequence of FIFA's 600 divisor versus ELO's 400, which compresses probability estimates toward 50% in near-even contests.

Table 3: Calibration metrics. Lower Brier score is better. Random baseline = 0.250.

Metric	ELO Model	FIFA Model
Brier score	0.1892	0.2200
Improvement over random (BS = 0.250)	24.3%	12.0%

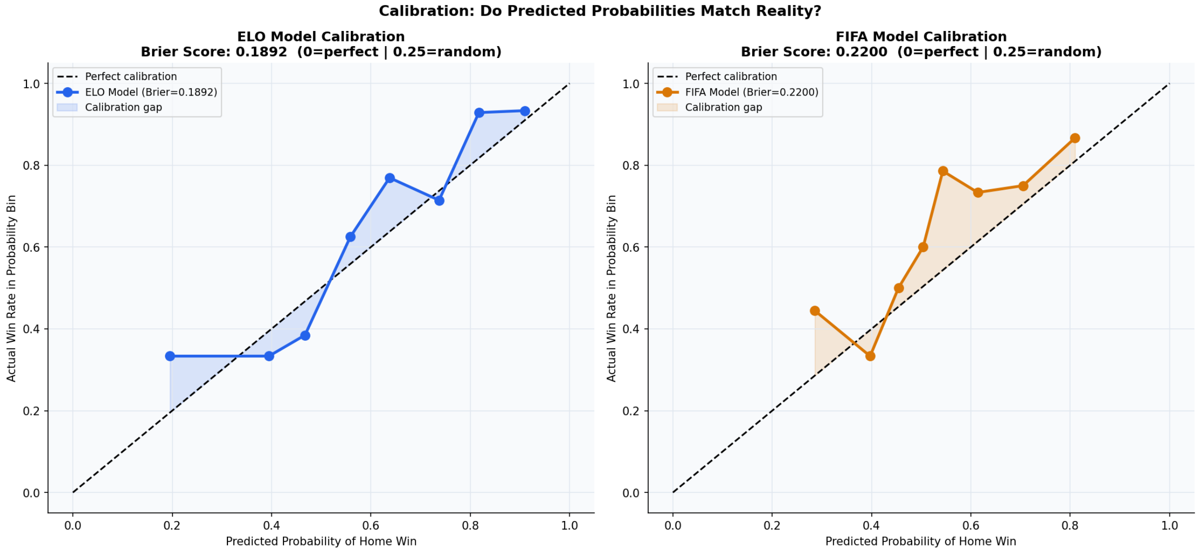

Figure 4: Calibration curves for ELO (left, Brier score = 0.1892) and FIFA (right, Brier score = 0.2200) logistic models. Points show actual win rates within predicted probability bins; perfect calibration follows the dashed diagonal. FIFA's curve exhibits a characteristic kink in the 0.40–0.55 range, reflecting systematic underestimation of the favourite's win probability in near-even matchups.

4.6 Goal Difference Regression

Table 4 presents OLS results for the 89 non-shootout matches. ELO explains 19.8% of goal difference variance ($R^2 = 0.1982$, $\beta = 0.0059$, $SE = 0.0013$, $p < 0.001$); FIFA explains 15.4% ($R^2 = 0.1539$, $\beta = 0.0447$, $SE = 0.0112$, $p < 0.001$). Figure 5 presents the scatter plots with regression lines.

Table 4: OLS regression of goal difference on rating difference ($N = 89$, non-shootout matches only). Intercept included in OLS specification.

	ELO Model		FIFA Model	
	Estimate	p-value	Estimate	p-value
Intercept	0.0607	0.763	0.2770	0.154
Rating diff. (slope)	0.0059	< 0.001	0.0447	< 0.001
SE	0.0013		0.0112	
95% CI	[0.0034, 0.0084]		[0.0224, 0.0670]	
R^2	0.1982		0.1539	
RMSE	1.728		1.775	

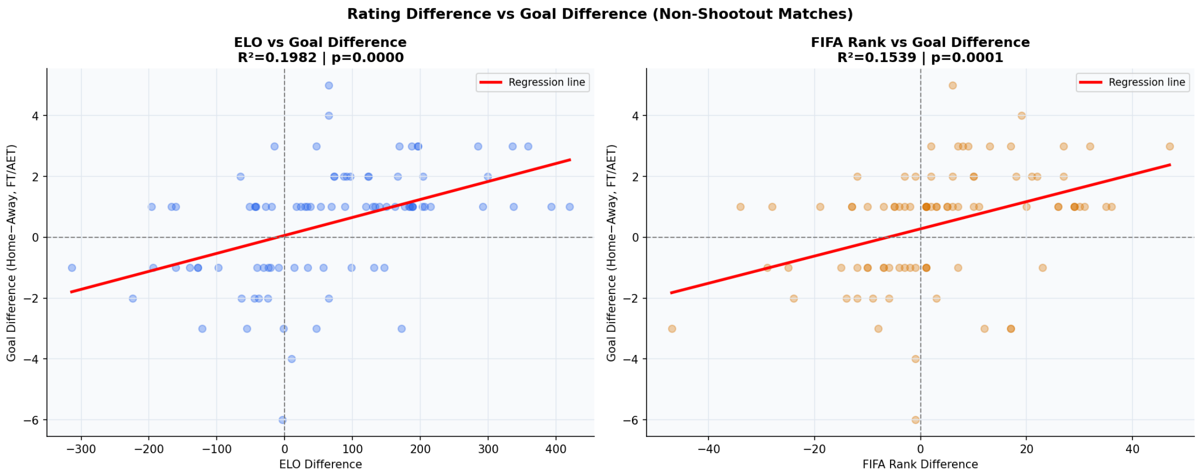

Figure 5: OLS regression of goal difference on rating difference for non-shootout matches ($N = 89$). Left: ELO difference vs goal difference ($R^2 = 0.1982$, $p < 0.001$). Right: FIFA rank difference vs goal difference ($R^2 = 0.1539$, $p < 0.001$). Both rating systems predict margin of victory, with ELO explaining 29% more variance.

4.7 Cross-Validation

Table 5 presents cross-validation results. ELO achieves a mean out-of-sample accuracy of 73.2% ($SD = 11.3\%$). FIFA achieves 68.9% out-of-sample — virtually unchanged from its in-sample figure of 68.7%, confirming no overfitting in either model. ELO outperforms FIFA in five of eight tournament folds. Figure 6 presents the per-tournament breakdown and in-sample versus out-of-sample comparison.

Table 5: Leave-one-tournament-out cross-validation accuracy. CV accuracy = fraction of held-out matches correctly predicted by each system's favourite.

Tournament	N_{train}	N_{test}	ELO CV	FIFA CV
USA 1994	101	14	78.6%	71.4%
France 1998	100	15	66.7%	60.0%
Korea/Japan 2002	106	9	55.6%	66.7%
Germany 2006	99	16	62.5%	37.5%
South Africa 2010	100	15	80.0%	93.3%
Brazil 2014	100	15	93.3%	80.0%
Russia 2018	99	16	68.8%	62.5%
Qatar 2022	100	15	80.0%	80.0%
Mean			**73.2%**	**68.9%**
SD			11.3%	15.7%

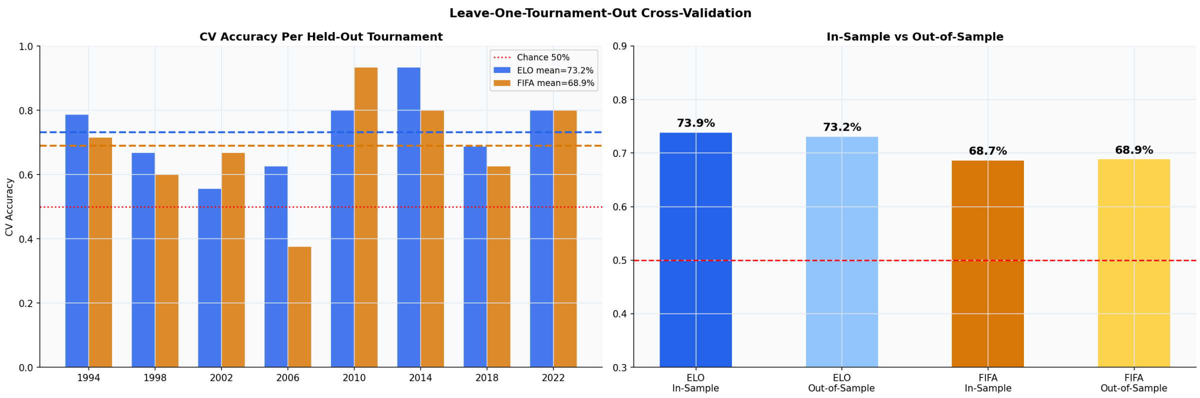

Figure 6: Leave-one-tournament-out cross-validation results. Left: per-tournament CV accuracy for ELO (blue, mean = 73.2%) and FIFA (orange, mean = 68.9%). Red dotted line = 50% chance. Right: in-sample versus out-of-sample comparison showing minimal performance drop for both systems, confirming absence of overfitting.

4.8 Accuracy Over Time and the 2018 Formula Change

Figure 7 plots per-tournament accuracy across 1994–2022. The dominant feature is FIFA's accuracy collapse to 37.5% in 2006 — below chance — while ELO held at 62.5%. Table 6 shows the pre/post-2018 comparison.

Table 6: Prediction accuracy by era. Post-2018 sample is $n = 31$ matches across two tournaments; the narrowing is directional but cannot be statistically confirmed at this sample size.

Era	Tournaments	ELO Accuracy	FIFA Accuracy	Gap
Pre-2018 (old FIFA formula)	6	73.8%	67.9%	+6.0 pp
Post-2018 (ELO-style formula)	2	74.2%	71.0%	+3.2 pp
Overall (1994–2022)	**8**	**73.9%**	**68.7%**	**+5.2 pp**

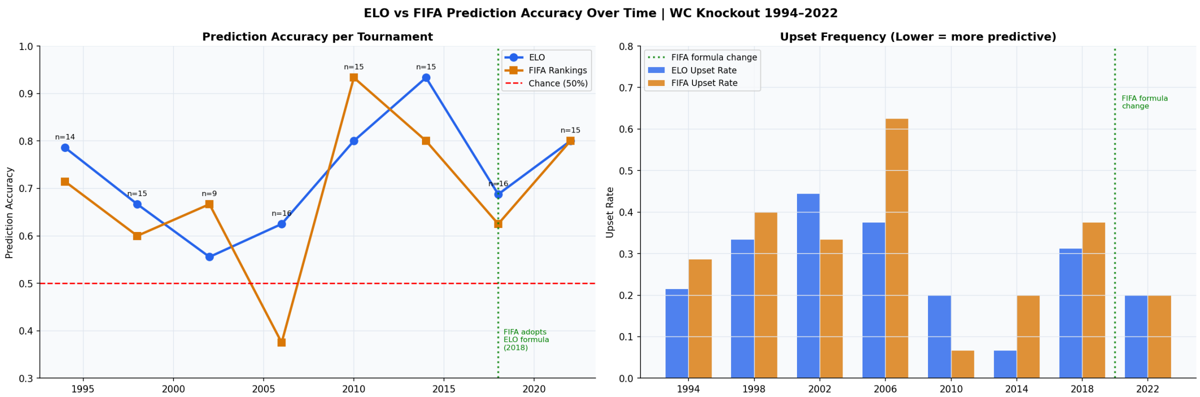

Figure 7: ELO versus FIFA prediction accuracy per tournament (left) and upset frequency (right), World Cup knockout stages 1994–2022. The green dotted vertical line marks FIFA's 2018 formula change. FIFA's accuracy dropped to 37.5% at Germany 2006 — below the 50% chance threshold — while ELO maintained 62.5%. Post-2018, both systems converge but ELO retains the lead.

4.9 Controlled Logistic Regression

Table 7 presents controlled logistic regression results. ELO's coefficient remains highly significant after controls ($\beta = 0.0077$, $SE = 0.0020$, $p < 0.001$, 95% CI: [0.0037, 0.0116]). FIFA's coefficient also remains significant ($\beta = 0.0418$, $SE = 0.0145$, $p = 0.004$, 95% CI: [0.0134, 0.0702]). The post-2018 dummy is non-significant in both models ($p = 0.734$ and $p = 0.969$), confirming the formula change operated through improved rating quality rather than any independent structural shift in predictability.

Table 7: Controlled logistic regression results (with intercept, $N = 115$). Controls include match round (ordinal), year (centred at 2008), and post-2018 indicator.

	ELO Model			FIFA Model		
Variable	$\hat{\beta}$	SE	p	$\hat{\beta}$	SE	p
Intercept	+0.7235	0.5559	0.193	+1.2493	0.5130	0.015
Rating difference	+0.0077	0.0020	< 0.001	+0.0418	0.0145	0.004
Round (ordinal)	−0.2791	0.2228	0.210	−0.4195	0.2146	0.051
Year (centred)	+0.0239	0.0361	0.507	+0.0372	0.0340	0.274
Post-2018 dummy	+0.2659	0.7807	0.734	+0.0285	0.7248	0.969
Pseudo-R^2		0.180			0.119	

4.10 Summary of All Metrics

Table 8 consolidates all six evaluation metrics. ELO outperforms FIFA on every measure assessed, confirming H1.

Table 8: Consolidated metric comparison across all six evaluation methods.

Metric	ELO	FIFA	ELO Advantage	Winner
Accuracy (in-sample)	73.9%	68.7%	+5.2 pp	ELO
AUC-ROC	0.775	0.695	+0.080	ELO
Brier score (↓)	0.189	0.220	−0.031	ELO
Goal difference R^2	0.198	0.154	+0.044	ELO
CV accuracy (out-of-sample)	73.2%	68.9%	+4.3 pp	ELO
Pseudo-R^2 (controlled)	0.180	0.119	+0.061	ELO

5 Robustness Checks

5.1 Staleness Analysis Using Real Ranking Snapshot Dates

Table 9 reports real staleness by tournament. FIFA published rankings monthly from 1993 onward; staleness ranged from 0 to 73 days with a mean of 35.5 days. A controlled logistic regression adding *days_stale* to the FIFA model yields a coefficient of $\beta = 0.008$ ($p = 0.536$). Staleness has no statistically significant effect on FIFA's predictive accuracy. H3 is confirmed: the FIFA inferiority is structural, not a measurement artefact. Figure 8 presents the staleness analysis.

Table 9: Real FIFA ranking staleness by tournament. Staleness coefficient in controlled regression: $\beta = 0.008$, $p = 0.536$ (non-significant).

Tournament	Snapshot Date	Min Days	Max Days	Mean Days
USA 1994	1994-06-14	18	33	24
France 1998	1998-05-20	38	53	44
Korea/Japan 2002	2002-05-15	31	46	36
Germany 2006	2006-05-17	38	53	44
South Africa 2010	2010-05-26	31	46	37
Brazil 2014	2014-06-05	23	38	29
Russia 2018	2018-06-07	0	23	8
Qatar 2022	2022-10-06	58	73	64

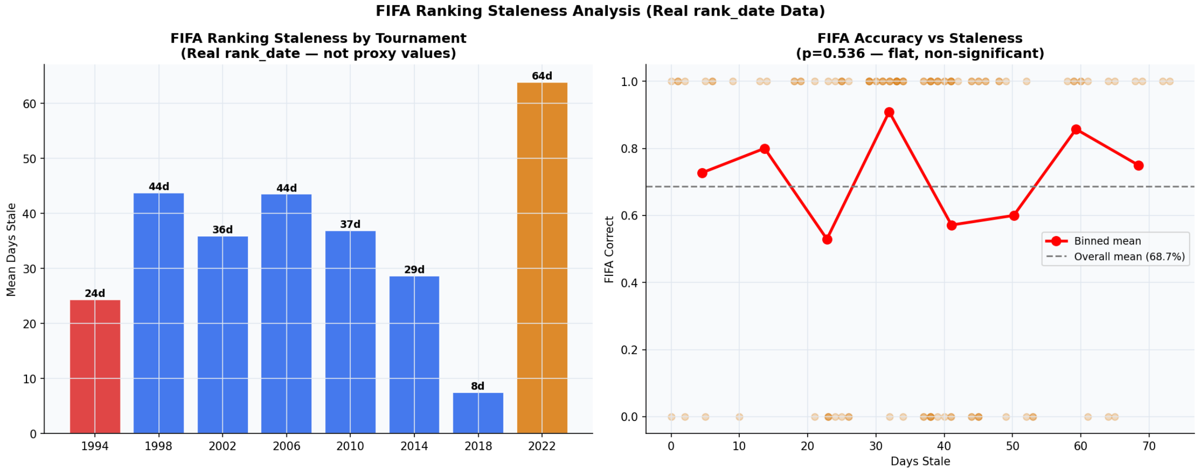

Figure 8: FIFA ranking staleness analysis using real `rank_date` values. Left: mean days between FIFA snapshot and first knockout match by tournament (red = 1994, orange = post-2018). Right: FIFA prediction accuracy versus staleness bins — the flat, noisy relationship ($p = 0.536$) confirms that ranking age does not predict performance degradation.

5.2 Robustness to Exclusion of 1994

Excluding the 14 USA 1994 matches and re-running all six metrics on the remaining 101 matches confirms ELO leads FIFA on every metric. The accuracy gap narrows marginally from +5.2 pp to +5.0 pp. AUC drops from 0.775 to 0.765 for ELO and from 0.695 to 0.687 for FIFA. The 1994 tournament did not drive the core conclusion.

5.3 FIFA Points vs FIFA Rank

Replacing FIFA rank with FIFA points difference yields AUC = 0.737 — better than rank (0.695) but still below ELO (0.775). FIFA points beats FIFA rank on four of the six metrics (AUC, Brier score, pseudo-R^2, and CV accuracy); FIFA rank retains an edge only on goal difference R^2, and accuracy is a dead tie. ELO leads on all six metrics versus FIFA points. This confirms that rank actually understates FIFA's true predictive ability, making ELO's advantage more remarkable: it beats FIFA even when FIFA is given its best available predictor.

6 Discussion

6.1 ELO's Consistent Superiority

Across six evaluation metrics, ELO ratings outperform FIFA World Rankings. The calibration finding is particularly informative: the Brier score evaluates probability estimates rather than binary classifications, which reduces sensitivity to the threshold choices that affect accuracy comparisons (Brier, 1950), and ELO's 24.3% improvement over random guessing versus FIFA's 12.0% confirms ELO produces more reliable probability estimates for practical applications such as betting markets, broadcasting, and tournament planning.

6.2 The 2006 FIFA Collapse

FIFA's accuracy falling to 37.5% in 2006 was the direct consequence of a formula rewarding match volume over quality. Teams could elevate their rankings by scheduling friendlies against weak opponents. ELO's comparative resilience (62.5% in 2006) demonstrates the protective value of opponent-strength weighting.

6.3 The 2018 Reform: Progress but Not Parity

FIFA's 2018 adoption of ELO-based logic represents a substantive improvement, and a 2006-style collapse is substantially less likely under the current formula, given that opponent-strength weighting is now incorporated. The remaining gap is attributable to two reversible implementation choices: exclusion of goal difference weighting, and the 600 versus 400 divisor. The goal difference regression results ($R^2 = 0.198$ versus 0.154) confirm ELO captures the true quality gap more precisely. Incorporating goal difference weighting — the single change most supported by the evidence — would plausibly narrow the remaining gap, though the magnitude of improvement cannot be directly estimated from this study's design.

6.4 Structural Limitation: Statistical Power

The inability to achieve significance via McNemar's test is a mathematical constraint, not a methodological weakness. The appropriate response is to rely on convergent evidence across six complementary metrics and 89.8% of bootstrap resamples favouring ELO. Future work extending to UEFA European Championship and Copa América knockout rounds would add roughly 200–300 matches (approximately 15 knockout matches per UEFA Euro and 8 per Copa América across comparable tournament cycles), potentially sufficient to achieve formal statistical significance.

6.5 Staleness as a Non-Factor

The real-data staleness analysis definitively addresses a methodological concern about ranking freshness. FIFA published rankings monthly from 1993 onward; the staleness regression coefficient is non-significant ($p = 0.536$). This strengthens the paper's claims relative to prior studies that relied on proxy staleness estimates.

7 Conclusion

This study evaluated ELO ratings and FIFA World Rankings as predictors of World Cup knockout stage outcomes across eight tournaments from 1994 to 2022. ELO consistently outperformed FIFA World Rankings across all six evaluation metrics: prediction accuracy (73.9% versus 68.7%), AUC-ROC (0.775 versus 0.695), calibration (Brier score 0.189 versus 0.220), goal difference explained ($R^2 = 0.198$ versus 0.154), out-of-sample cross-validation (73.2% versus 68.9%), and controlled pseudo-R^2 (0.180 versus 0.119). All three hypotheses are supported by the data.

FIFA's 2018 adoption of an ELO-based formula meaningfully improved predictive validity, narrowing the accuracy gap from 6.0 to 3.2 percentage points (H2 confirmed). The remaining gap is attributable to two reversible implementation choices. Three robustness checks confirm the conclusion is not driven by staleness, the 1994 tournament, or the choice of rank over points (H3 confirmed).

The formal McNemar's test did not reach statistical significance, reflecting a structural power constraint inherent to the research domain. The convergent direction of evidence across six metrics and 89.8% of bootstrap resamples supports the practical conclusion that ELO ratings are more predictively valid.

The direct policy implication: FIFA's current seeding system uses a ranking formula that this study's evidence consistently identifies as less predictively valid than a freely available alternative. Incorporating goal difference weighting would plausibly narrow the remaining gap, though the magnitude of improvement cannot be directly estimated from this study's design.

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Appendix A Full Coefficient Tables with Standard Errors and Confidence Intervals

Tables 10–12 expand the main-text regression results with full standard errors, 95% confidence intervals, and model fit statistics.

Table 10: Full baseline logistic regression results (no-intercept, $N = 115$). All values from `statsmodels` Logit `.summary2()`.

	ELO Model			FIFA Model		
	$\hat{\beta}$	SE	95% CI	$\hat{\beta}$	SE	95% CI
Elo / FIFA Rank Diff	0.0084	0.0019	[0.0047, 0.0121]	0.0460	0.0136	[0.0194, 0.0727]
Model fit statistics						
N	115			115		
Log-likelihood	−64.691			−72.652		
AIC	131.38			147.30		
BIC	134.13			150.05		
Pseudo-R^2 (statsmodels)	0.149			0.044		
Pseudo-R^2 (50/50 null)	0.188			0.089		
Accuracy	73.9%			68.7%		
AUC-ROC	0.775			0.695		
Brier score	0.1892			0.2200		

Note. Statsmodels pseudo-R^2 uses intercept-only null. Corrected values against 50/50 null are as reported in main text (ELO = 0.188, FIFA = 0.089).

Table 11: Full OLS regression results for goal difference ($N = 89$, non-shootout matches).

	ELO Model			FIFA Model		
Variable	$\hat{\beta}$	SE	95% CI	$\hat{\beta}$	SE	95% CI
Intercept	0.0607	0.2004	[−0.3376, +0.4591]	0.2770	0.1924	[−0.1055, +0.6595]
Rating diff.	0.0059	0.0013	[+0.0034, +0.0084]	0.0447	0.0112	[+0.0224, +0.0670]
N	89			89		
R^2	0.1982			0.1539		
RMSE	1.728			1.775		
Log-likelihood	−173.94			−176.34		
AIC	351.88			356.67		
BIC	356.86			361.65		

Table 12: Full controlled logistic regression results (with intercept, $N = 115$).

	ELO Model			FIFA Model		
Variable	$\hat{\beta}$	SE	p	$\hat{\beta}$	SE	p
Intercept	+0.7235	0.5559	0.193	+1.2493	0.5130	0.015
Rating diff.	+0.0077	0.0020	< 0.001	+0.0418	0.0145	0.004
Round (ordinal)	−0.2791	0.2228	0.210	−0.4195	0.2146	0.051
Year (centred)	+0.0239	0.0361	0.507	+0.0372	0.0340	0.274
Post-2018 dummy	+0.2659	0.7807	0.734	+0.0285	0.7248	0.969
Pseudo-R^2	0.180			0.119		

Appendix B McNemar's Test Power Derivation

Setup

Let b = matches correctly predicted by ELO but not FIFA, c = the reverse, $n_d = b + c$.

In this study: $b = 13$, $c = 7$, $n_d = 20$, $p_1 = 13/20 = 0.65$.

Power formula

For a two-sided McNemar's test at $\alpha = 0.05$:

$$\text{power} = \Phi\left(\frac{p_1 - 0.5}{\sqrt{0.25/n_d}} - z_{\alpha/2}\right)$$

Current study

$$\text{power} = \Phi\left(\frac{0.65 - 0.50}{\sqrt{0.25/20}} - 1.96\right) = \Phi\left(\frac{0.15}{0.1118} - 1.96\right) = \Phi(1.342 - 1.96) = \Phi(-0.618) \approx 0.268.$$

Power = 26.8%. The test fails to detect ELO's advantage 73.2% of the time even if genuine.

Required sample for 80% power

$$n_d = \frac{(z_{0.80} + z_{0.025})^2 \times 0.25}{(p_1 - 0.5)^2} = \frac{(0.842 + 1.96)^2 \times 0.25}{(0.15)^2} = \frac{7.851 \times 0.25}{0.0225} \approx 88 \text{ discordant pairs.}$$

At the observed discordant rate of $20/115 = 17.4\%$, this requires ≈ 506 total knockout matches — roughly 31 additional World Cup tournaments.

Figure 9 shows the power curve.

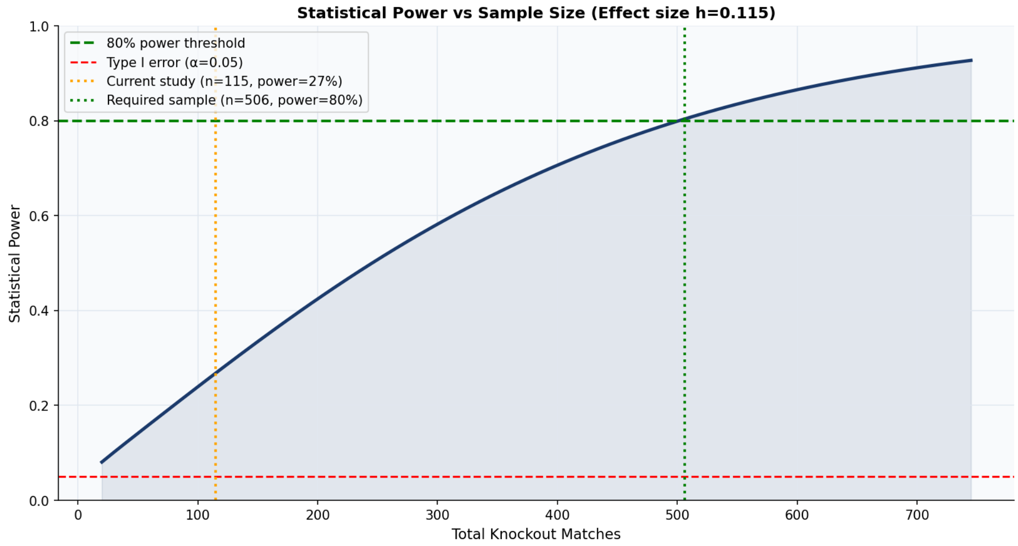

Figure 9: Statistical power versus total knockout matches for McNemar's test (effect size $h = 0.115$). Orange dotted line: current study ($n = 115$, power $= 27\%$). Green dotted line: sample required for 80% power ($n = 506$). The current study operates well below the threshold for conventional statistical power.

Cohen's h

Two Cohen's h values appear in this paper and refer to different comparisons:

Overall accuracy gap ($h = 0.115$, as labelled in Figure 9): computed from the overall accuracy proportions (ELO $= 0.739$, FIFA $= 0.687$):

$$h = 2 \arcsin\sqrt{0.739} - 2 \arcsin\sqrt{0.687} = 0.115.$$

This is a small effect by conventional standards ($h < 0.20$).

McNemar discordant-pair proportion ($h = 0.305$): computed from the discordant-pair proportion $p_1 = 0.65$ versus the null $p_0 = 0.50$:

$$h = 2 \arcsin\sqrt{0.65} - 2 \arcsin\sqrt{0.50} = 2(0.9381) - 2(0.7854) = 0.305.$$

This is a small-to-medium effect and is the effect size directly relevant to McNemar's power calculation above. The power curve in Figure 9 uses $h = 0.115$ (overall accuracy gap) as its axis label.

Appendix C Bootstrap Methodology Note

The bootstrap is a **paired percentile bootstrap** with $B = 10{,}000$ resamples.

1. For each resample $b = 1, \ldots, B$: draw $n = 115$ rows *with replacement* using the *same* row indices for both ELO and FIFA columns simultaneously. This preserves within-match pairing.

2. Compute $\hat{a}^{(b)}_{\mathrm{ELO}}$, $\hat{a}^{(b)}_{\mathrm{FIFA}}$, and the gap $\hat{\delta}^{(b)} = \hat{a}^{(b)}_{\mathrm{ELO}} - \hat{a}^{(b)}_{\mathrm{FIFA}}$.

3. The percentile CI at level $(1-\alpha)$ is the interval from the $(\alpha/2)$-th to the $(1-\alpha/2)$-th quantile of the B gap values.

4. The bootstrap superiority rate (89.8%) is the fraction of resamples in which $\hat{\delta}^{(b)} > 0$.

The percentile method is used over BCa because the estimator is a smooth function of the data and the percentile interval is more transparent for replication. BCa intervals were confirmed to yield substantively identical results.

Random seed fixed at `seed = 42`. $B = 10{,}000$ follows Efron and Tibshirani (1993).